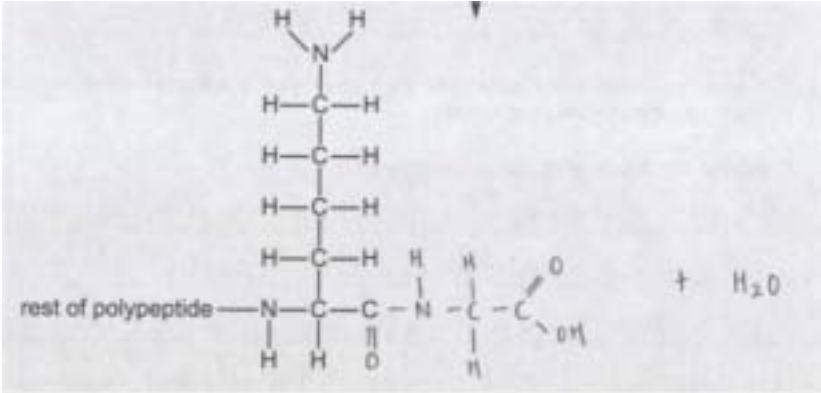


|           |                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(a)(i)   | <p><b>I</b> <i>presence or absence of water</i><br/>         diagram shows bond between C (of lysine) and N (of glycine) <b>and</b><br/>         double bond to O shown correctly ;<br/>         glycine drawn correctly ;<br/> <b>R</b> if R is used instead of H on glycine</p>                                                                | <b>2</b> |
| 5(a)(ii)  | condensation ; <b>A</b> dehydration (reaction)                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 5(a)(iii) | <p><b>I</b> high tensile strength<br/> <i>any two from:</i><br/>         allows close packing of, triple helix / three polypeptides ; <b>A</b> chains<br/> <b>I</b> tight coiling <b>A</b> 'binds more tightly'<br/>         glycine, has smallest R group / R group of glycine is H / is smallest amino acid ; <b>A</b> glycine is small<br/>         glycine, is <u>every third</u> amino acid in the, polypeptide / chain ;</p> | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(b)     | <p><b>I</b> collagen is insoluble / collagen is strong<br/> any <b>four</b> from:</p> <p><b>1</b> collagen, has high tensile strength / does not stretch / AW ;<br/> <b>A</b> withstands large pulling forces</p> <p><b>2</b> any ref. to use of collagen in the body ;<br/> e.g. (walls of) arteries or veins / tendons / cartilage / skin / basement membranes / around alveoli / bones / teeth<br/> <b>I</b> hair / <b>R</b> if any ref. to elastic(icity)</p> <p><i>three from</i></p> <p><b>3</b> triple helix has many hydrogen bonds between, polypeptides / chains ;</p> <p><i>mp4, 5, 6 and 7 – must be clear that answers are about molecules, but accept triple helix for molecule</i></p> <p><b>4</b> collagen molecules form, fibrils / fibres ;</p> <p><b>5</b> strong / covalent, (cross) links between molecules ;</p> <p><b>6</b> (ends of) molecules (in fibril / fibre) are staggered ;</p> <p><b>7</b> AVP ; e.g. (many) collagen molecules lie parallel</p> | <b>4</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(c)     | <p><i>induced fit</i></p> <p><b>1</b> active site changes shape ;<br/> <b>A</b> active site moulds round substrate<br/> <b>A</b> 'active site changes to fit the substrate (more closely)'</p> <p><b>2</b> (so) active site becomes (fully) <u>complementary</u> shape to collagen ;</p> <p><i>any four from</i></p> <p><b>3</b> formation of, <u>enzyme-substrate complex</u> / <u>ESC</u> ;</p> <p><b>4</b> lowering of activation energy ;</p> <p><b>5</b> breakage of peptide bond ;</p> <p><b>6</b> active site returns to, pre-ESC / original, shape <b>and</b> can be reused ;</p> <p><b>7</b> AVP ;<br/> e.g. <i>ref. to</i>, binding site / catalytic site (of active site)<br/> e.g. suggestion of how activation energy lowered, e.g. strain put on bond / alternative pathway</p> | <b>5</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(d)     | <p><i>description</i></p> <p><b>1</b> general description of effect of pH on activity ;<br/> e.g. as pH increases the activity increases and then decreases<br/> e.g. as pH increases activity reaches a peak<br/> e.g. use of data with unit for relative activity to describe increase and decrease (see next page)</p> <p><b>2</b> <u>2 optimum</u> pH is 7 ;</p> <p><i>explanation</i><br/> any <b>three</b> from:</p> <p><b>3</b> partial denaturation in, acid <b>and</b> alkaline conditions / AW ;<br/> <b>A</b> 'starts to denature' as alternative to partial denaturation</p> <p><i>small changes either side of optimum</i></p> <p><b>4</b> (ionisable) R-groups in, active / catalytic, site affected ;</p> <p><i>large changes either side of optimum</i></p> <p><b>5</b> <u>hydrogen bonds</u> / <u>ionic bonds</u>, break / disrupted ;</p> <p><b>6</b> active site (shape) / (enzyme) tertiary structure, changes, so substrate / collagen, no longer fits into active site / fewer ESC complexes formed ;<br/> <b>A</b> fits less well <i>in context of partial denaturation</i></p> <p><b>7</b> AVP ; e.g. detail of R groups</p> | <b>4</b> |